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СОРБЦИЯ ОРГАНИЧЕСКИХ МИКРОЗАГРЯЗНЕНИЙ ЛИПОСОМАМИ И ХЕМОИНФОРМАЦИОННЫЙ АНАЛИЗ МЕХАНИЗМА ИХ ВЗАИМОДЕЙСТВИЯ

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Коэффициенты распределения между липосомами и водой ($logK_{lip/w}$) обеспечивают аппроксимацию сродства к биологическим мембранам по сравнению с традиционным распределением растворенного вещества в системе октанол-вода. Хотя количественные соотношения «структура–свойство» (QSPR) для предсказания $logK_{lip/w}$ достигают высокой статистической точности, интерпретация механизма для оценки экологического риска остается ограниченной. В настоящей работе мы переосмысливаем дескрипторы молекулярных фрагментов, чтобы выявить молекулярные механизмы, управляющие распределением в мембранах. Анализ 306 химически разнообразных органических соединений, включая нейтральные вещества, ионизируемые фармацевтические препараты, галогенированные ароматические соединения и перфторированные вещества, позволил идентифицировать 10 интерпретируемых фрагментов, коэффициенты которых напрямую соотносятся с биофизикой мембран: (1) π-стэкинг ароматических систем с липидными хвостами доминирует в управлении распределением, тогда как галогенное связывание вносит меньший, но статистически значимый вклад; (2) эфирные и карбонильные атомы кислорода создают выраженный энергетический барьер при распределении вследствие потери благоприятных водородных связей с водной фазой при переходе в гидрофобное ядро бислоя; (3) увеличение молекулярного размера масштабирует гидрофобные взаимодействия, при этом оптимальное соответствие алкильных цепей (C8–C12) толщине гидрофобного ядра максимизирует сродство к мембране. Фрагментный анализ объясняет аномалии, недоступные метрикам на основе объемной гидрофобности. Фрагментный подход обеспечивает переход от статистического предсказания к раскрытию физико-химических механизмов распределения, что позволяет формулировать количественные правила «зеленого» молекулярного дизайна для снижения мембранного сродства органических микрозагрязнений.

Ключевые слова: распределение липосомы/вода; сорбция, QSPR; фрагментные дескрипторы; биофизика мембран; галогенное связывание; биоаккумуляция; интерпретация механизма

SORPTION OF ORGANIC MICROPOLLUTANTS BY LIPOSOMES AND CHEMOINFORMATICS ANALYSIS OF THEIR INTERACTION MECHANISM

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Liposome–water partition coefficients (log Klip/w) provide a more accurate approximation of affinity for biological membranes compared to the traditional octanol–water distribution system. Although quantitative structure–property relationships (QSPR) for predicting log Klip/w achieve high statistical accuracy, mechanistic interpretation for environmental risk assessment remains limited. In this work, we reinterpret molecular fragment descriptors to elucidate the molecular mechanisms governing membrane distribution. Analysis of 306 chemically diverse organic compounds—including neutral substances, ionizable pharmaceuticals, halogenated aromatics, and perfluorinated compounds—enabled identification of 10 interpretable fragments whose coefficients directly correlate with membrane biophysics: (1) π-stacking of aromatic systems with lipid tails dominates the regulation of distribution, whereas halogen bonding contributes a smaller but statistically significant effect; (2) ether and carbonyl oxygen atoms create a pronounced energetic barrier to partitioning due to the loss of favorable hydrogen bonds with the aqueous phase upon transfer into the hydrophobic core of the bilayer; (3) increasing molecular size scales with hydrophobic interactions, while optimal matching of alkyl chain length (C8–C12) to the thickness of the hydrophobic core maximizes membrane affinity. Fragment-based analysis explains anomalies inaccessible to bulk hydrophobicity metrics. The fragment approach enables a shift from statistical prediction toward elucidation of physicochemical distribution mechanisms, allowing formulation of quantitative rules for "green" molecular design aimed at reducing membrane affinity of organic micropollutants.

Keywords: liposome–water distribution; membrane sorption; quantitative structure–property relationships (QSPR); fragment descriptors; membrane biophysics; halogen bonding; bioaccumulation; interpretation of interaction mechanisms

Table S1: Dataset of liposomes and chemicals liposome-water partitioning (2019-S_Lin-pH7_4_decoded.csv)

ID	CAS	Name	logDlip/w-(pH 7.40)
	000110-54-3	n-hexane	3.91
2	000142-82-5	n-heptane	4.55
3	000111-65-9	n-octane	4.67
4	000540-84-1	2,2,4-trimethylpentane	4.61
5	000110-82-7	cyclohexane	3.27
6	000056-23-5	tetrachloromethane	2.61
7	000079-01-6	trichloroethene	2.43
8	000127-18-4	tetrachloroethene	3.08
9	000075-25-2	tribromomethane	2.33
10	000142-96-1	di-n-butyl ether	2.78
11	000693-65-2	di-n-pentyl ether	3.77
12	000067-64-1	acetone	0.1
13	000111-13-7	2-octanone	2.42
14	000821-55-6	2-nonanone	2.83
15	000693-54-9	2-decanone	3.16
16	000120-92-3	cyclopentanone	0.3
17	000108-94-1	cyclohexanone	0.54
18	000502-42-1	cycloheptanone	0.98
19	000502-49-8	cyclooctanone	1.4

20	000504-02-9	1,3-cyclohexanedione	0.49
21	000141-78-6	ethyl acetate	0.46
22	000109-60-4	propyl acetate	1.01
23	000105-53-3	diethyl malonate	0.5
24	000141-28-6	diethyl adipate	1.66
25	000141-03-7	dibutyl succinate	2.4
26	000541-35-5	butyramide	-0.2
27	000057-13-6	urea	-0.59
28	013214-66-9	4-phenylbutylamine	2.12
29	000067-56-1	methanol	-0.53
30	000064-17-5	ethanol	-0.26
31	000071-23-8	1-propanol	0.17
32	000067-63-0	2-propanol	-0.04
33	000071-36-3	1-butanol	0.51
34	000075-65-0	tert-butanol	0.16
35	000071-41-0	1-pentanol	1.08
36	000584-02-1	3-pentanol	1
37	000111-27-3	1-hexanol	1.88
38	000111-70-6	1-heptanol	2.38
39	000589-55-9	4-heptanol	1.7
40	027522-11-8	2-ethyl-1-pentanol	1.81
41	000600-36-2	2,4-dimethyl-3-pentanol	1.71
42	000111-87-5	1-octanol	2.66
43	000096-41-3	cyclopentanol	0.52
44	000108-93-0	cyclohexanol	1.02
45	000502-41-0	cycloheptanol	1.51
46	000696-71-9	cyclooctanol	2.09
47	000928-97-2	trans-3-hexen-1-ol	1.23
48	000928-95-0	trans-2-hexen-1-ol	1.36
49	000928-96-1	cis-3-hexen-1-ol	1.2
50	000821-41-0	5-hexen-1-ol	1.2
51	004938-52-7	1-hepten-3-ol	1.61
52	000928-90-5	5-hexyn-1-ol	0.78
53	000107-21-1	ethylene glycol	-0.79
54	005343-92-0	1,2-pentanediol	0.26
55	000111-29-5	1,5-pentanediol	-0.7
56	000625-69-4	2,4-pentanediol	-0.3
57	006920-22-5	1,2-hexanediol	0.81
58	000928-40-5	1,5-hexanediol	-0.22
59	000629-11-8	1,6-hexanediol	-0.1
60	000629-30-1	1,7-heptanediol	0.32
61	001117-86-8	1,2-octanediol	1.93
62	000629-41-4	1,8-octanediol	0.81
63	001460-57-7	trans-1,2-cyclohexanediol	0.23
64	042565-22-0	trans-1,2-cyclooctanediol	1.11
65	003031-66-1	3-hexyn-2,5-diol	-0.22
66	000056-81-5	glycerol	-1.04
67	000149-32-6	erythritol	-1.24

68	000111-76-2	2-butoxyethanol	0.6
69	000100-51-6	benzyl alcohol	1.14
70	002270-20-4	5-phenylvaleric acid	1.66
71	000071-43-2	benzene	2.1
72	000106-42-3	p-xylene	2.98
73	000108-90-7	chlorobenzene	2.81
74	000095-50-1	1,2-dichlorobenzene	3.64
75	000541-73-1	1,3-dichlorobenzene	3.71
76	000106-46-7	1,4-dichlorobenzene	3.57
77	000087-61-6	1,2,3-trichlorobenzene	4.22
78	000120-82-1	1,2,4-trichlorobenzene	4.2
79	000108-70-3	1,3,5-trichlorobenzene	4.09
80	000634-90-2	1,2,3,5-tetrachlorobenzene	4.63
81	000095-94-3	1,2,4,5-tetrachlorobenzene	4.59
82	000608-93-5	pentachlorobenzene	5.1
83	000118-74-1	hexachlorobenzene	5.75
84	006639-30-1	2,4,5-trichlorotoluene	4.72
85	000106-37-6	1,4-dibromobenzene	4.3
86	000626-39-1	1,3,5-tribromobenzene	5.21
87	000087-82-1	hexabromobenzene	5.64
88	034883-41-5	3,5-dichlorobiphenyl (PCB 14)	4.52
89	002050-68-2	4,4'-bichlorobiphenyl (PCB 15)	4.78
90	015862-07-4	2,4,5-trichlorobiphenyl (PCB 29)	5.16
91	035693-92-6	2,4,6-trichlorobiphenyl (PCB 30)	4.71
92	038444-93-8	2,2',3,3'-tetrachlorobiphenyl (PCB 40)	5.4
93	041464-40-8	2,2',4,5'-tetrachlorobiphenyl (PCB 49)	5.55
94	062796-65-0	2,2',4,6-tetrachlorobiphenyl (PCB 50)	5.92
95	035693-99-3	2,2',5,5'-tetrachlorobiphenyl (PCB 52)	5.94
96	033284-53-6	2,3,4,5-tetrachlorobiphenyl (PCB 61)	5.44
97	070362-49-1	3,3',4,5-tetrachlorobiphenyl (PCB 78)	6.53
98	060145-21-3	2,2',4,5',6-pentachlorobiphenyl (PCB 103)	6.32
99	056558-16-8	2,2',4,6,6'-pentachlorobiphenyl (PCB 104)	6.13
100	018259-05-7	2,3,4,5,6-pentachlorobiphenyl (PCB 116)	5.9
101	038380-07-3	2,2',3,3',4,4'-hexachlorobiphenyl (PCB 128)	6.09
102	038411-22-2	2,2',3,3',6,6'-hexachlorobiphenyl (PCB 136)	6.2
103	033979-03-2	2,2',4,4',6,6'-hexachlorobiphenyl (PCB 155)	5.98
104	035065-29-3	2,2',3,4,4',5,5'-heptachlorobiphenyl (PCB 180)	6.33
105	060145-23-5	2,2',3,4,4',5,6'-heptachlorobiphenyl (PCB 182)	6.83
106	068194-17-2	2,2',3,3',4,5,5',6-octachlorobiphenyl (PCB 198)	6.18
107	002051-24-3	decachlorobiphenyl (PCB 209)	6.17
108	000092-86-4	4,4'-dibromobiphenyl (PBB 15)	6.01
109	059080-33-0	2,4,6-tribromobiphenyl (PBB 30)	6
110	059080-37-4	2,2',5,5'-tetrabromobiphenyl (PBB 52)	6.05
111	059261-08-4	2,2',4,4',6,6'-hexabromobiphenyl (PBB 155)	5.96
112	002234-13-1	octachloronaphthalene	5.5
113	033857-26-0	2,7-dichlorodibenzo-p-dioxin	5.9
114	039227-58-2	1,2,4-trichlorodibenzo-p-dioxin	5.58
115	030746-58-8	1,2,3,4-tetrachlorodibenzo-p-dioxin	5.21

116	001746-01-6	2,3,7,8-tetrachlorodibenzo-p-dioxin	6.9
117	040321-76-4	1,2,3,7,8-pentachlorodibenzo-p-dioxin	7.07
118	039227-28-6	1,2,3,4,7,8-hexachlorodibenzo-p-dioxin	7.15
119	003268-87-9	octachlorodibenzo-p-dioxin	4.96
120	000085-01-8	phenanthrene	4.95
121	000120-12-7	anthracene	5.21
122	000206-44-0	fluoranthene	5.58
123	000129-00-0	pyrene	5.71
124	000056-55-3	benzo[a]anthracene	6.44
125	000218-01-9	chrysene	6.4
126	000192-97-2	benzo[e]pyrene	7.08
127	000205-99-2	benzo[b]fluoranthene	7.11
128	000207-08-9	benzo[k]fluoranthene	7.13
129	000050-32-8	benzo[a]pyrene	7.15
130	000191-24-2	benzo[ghi]perylene	7.66
131	000053-70-3	dibenz[a,h]anthracene	7.72
132	000215-58-7	dibenz[a,c]anthracene	7.49
133	000193-39-5	indeno[1,2,3-cd]pyrene	7.86
134	000098-95-3	nitrobenzene	2.01
135	000088-72-2	2-nitrotoluene	2.41
136	000091-22-5	quinoline	1.67
137	000084-66-2	diethylphthalate	1.77
138	000084-74-2	dibutylphthalate	3.87
139	000085-68-7	benzylbutylphthalate	4.3
140	042087-80-9	methyl-4-chloro-2-nitrobenzoate	2.45
141	000108-95-2	phenol	1.96
142	000095-48-7	2-methylphenol	2.45
143	000108-39-4	3-methylphenol	2.34
144	000106-44-5	4-methylphenol	2.35
145	000090-00-6	2-ethylphenol	2.81
146	000123-07-9	4-ethylphenol	2.79
147	000576-26-1	2,6-dimethylphenol	2.47
148	000644-35-9	2-n-propylphenol	3.13
149	000645-56-7	4-n-propylphenol	2.92
150	000099-89-8	4-isopropylphenol	3.25
151	000527-54-8	3,4,5-trimethylphenol	2.66
152	001638-22-8	4-n-butylphenol	3.13
153	000089-72-5	2-sec-butylphenol	3.47
154	000099-71-8	4-sec-butylphenol	3.26
155	000088-18-6	2-tert-butylphenol	3.51
156	000585-34-2	3-tert-butylphenol	3.25
157	000098-54-4	4-tert-butylphenol	3.48
158	001006-59-3	2,6-diethylphenol	2.73
159	014938-35-3	4-n-pentylphenol	4.31
160	000080-46-6	4-tert-amylphenol	3.54
161	000140-66-9	4-tert-octylphenol	5.39
162	000104-40-5	4-n-nonylphenol	5.57
163	001745-81-9	2-allylphenol	3.06

164	000098-17-9	3-trifluoromethylphenol	3.25
165	000090-43-7	2-phenylphenol	3.43
166	000092-69-3	4-phenylphenol	3.52
167	000080-05-7	bisphenol A	3.92
168	000084-16-2	meso-hexestrol	4.37
169	000084-17-3	dienestrol	4.83
170	000056-53-1	diethylstilbestrol	4.76
171	000108-43-0	3-chlorophenol	2.78
172	000106-48-9	4-chlorophenol	2.72
173	000059-50-7	3-methyl-4-chlorophenol	3.32
174	000120-83-2	2,4-dichlorophenol	3.15
175	000087-65-0	2,6-dichlorophenol	1.57
176	000095-77-2	3,4-dichlorophenol	3.67
177	000095-95-4	2,4,5-trichlorophenol	3.22
178	000088-06-2	2,4,6-trichlorophenol	2.57
179	000058-90-2	2,3,4,6-tetrachlorophenol	3.46
180	000087-86-5	pentachlorophenol	4.32
181	000371-41-5	4-fluorophenol	2.2
182	000106-41-2	4-bromophenol	2.39
183	013979-81-2	3,5-dibromo-4-methylphenol	4.44
184	000540-38-5	4-iodophenol	2.57
185	000088-75-5	2-nitrophenol	0.86
186	000554-84-7	3-nitrophenol	2.52
187	000051-28-5	2,4-dinitrophenol	1.9
188	000534-52-1	2-methyl-4,6-dinitrophenol	2.35
189	004099-71-2	2-ethyl-4,6-dinitrophenol	3.02
190	000088-85-7	2-sec-butyl-4,6-dinitrophenol	3.35
191	001689-84-5	3,5-dibromo-4-hydroxybenzonitrile	2.1
192	000062-53-3	aniline	1.63
193	000095-64-7	3,4-dimethylaniline	2.11
194	000636-30-6	2,4,5-trichloroaniline	4.16
195	000088-05-1	2,4,6-trimethylaniline	2.38
196	003481-20-7	2,3,5,6-tetrachloroaniline	4.6
197	000527-20-8	pentachloroaniline	4.69
198	000099-09-2	3-nitroaniline	2.17
199	005372-81-6	dimethyl-2-amino-p-phthalate	2.53
200	000121-69-7	N,N-dimethylaniline	2.33
201	000118-92-3	anthranilic acid	0.13
202	000053-16-7	estrone	3.6
203	000050-28-2	17?-estradiol	3.33
204	000050-27-1	estriol	1.96
205	000057-63-6	17?-ethynylestradiol	3.96
206	000057-83-0	progesterone	3.28
207	000439-14-5	diazepam	2.99
208	000739-71-9	trimipramine	3.31
209	000081-81-2	warfarin	1.43
210	088150-42-9	amlodipine	3.75
211	000525-66-6	propranolol	2.74

212	054910-89-3	fluoxetine	4.03
213	126924-38-7	norfluoxetine	4.07
214	000137-58-6	lidocaine	1.4
215	000059-46-1	procaine	0.77
216	000094-24-6	tetracaine	2.21
217	015307-86-5	diclofenac	4.45
218	015687-27-1	ibuprofen	1.81
219	000069-72-7	salicylic acid	0.97
220	000050-78-2	acetylsalicylic acid	1.6
221	059277-89-3	acyclovir	1.76
222	000315-30-0	allopurinol	2.6
223	002609-46-3	amiloride	1.71
224	029122-68-7	atenolol	1.01
225	076824-35-6	famotidine	1.75
226	000054-31-9	furosemide	1.9
227	022916-47-8	miconazole	3.55
228	075438-57-2	moxonidine	1.59
229	076963-41-2	nizatidine	2.82
230	132539-06-1	olanzapine	3.29
231	091161-71-6	terbinafine	3.1
232	014293-44-8	xipamide	1.79
233	030516-87-1	zidovudine	2.4
234	043200-80-2	zopiclone	1.7
235	000699-04-7	(p-methylbenzyl)methylamine	2.54
236	039190-96-0	(p-methylbenzyl)propylamine	2.11
237	170303-38-5	(p-methylbenzyl)pentylamine	1.84
238	215177-24-5	(p-methylbenzyl)heptylamine	2.71
239	000095-68-1	3,4-dimethylaniline	2.11
240	037517-30-9	acebutolol	0.66
241	013655-52-2	alprenolol	2.17
242	023284-25-5	bupranolol	2.49
243	083881-51-0	cetirizine	3.2
244	000050-53-3	chlorpromazine	3.43
245	000068-88-2	hydroxyzine	2.98
246	312753-06-3	indacaterol	3.56
247	036894-69-6	labetalol	2.32
248	000119-27-7	2,4-dinitroanisole	1.6
249	000121-14-2	2,4-dinitrotoluene	2.41
250	000099-35-4	2,4,6-trinitrobenzene	1.73
251	000118-96-7	trinitrotoluene	2.18
252	042200-33-9	nadolol	0.95
253	083891-03-6	norfluoxetine	3.84
254	006452-71-7	oxprenolol	1.51
255	016183-21-4	(p-methylbenzyl)butylamine	1.55
256	039099-13-3	(p-methylbenzyl)ethylamine	2.27
257	215177-23-4	(p-methylbenzyl)hexylamine	2.44
258	013523-86-9	pindolol	1.4
259	000130-95-0	quinine	2.2

260	089365-50-4	salmeterol	3.67
261	018198-39-5	Tetraphenyl-phosphonium	1.37
262	002933-94-0	toliprolol	1.49
263	004901-51-3	2,3,4,5-tetrachlorophenol	3.77
264	002131-41-1	1,4,5-Trimethylnaphthalene	5.11
265	017057-91-9	2,4,5-Trimethylnaphthalene	4.88
266	004569-45-3	9,9-Dimethylfluorene	5.01
267	000832-71-3	3-Methylphenanthrene	5.24
268	001576-67-6	3,6-Dimethylphenanthrene	5.58
269	000779-02-2	9-Methylanthracene	5.52
270	000781-43-1	9,10-Dimethylanthracene	5.57
271	002381-21-7	1-Methylpyrene	5.76
272	002541-69-7	7-Methylbenz[a]anthracene	6
273	000057-97-6	7,12-Dimethylbenz[a]anthracene	6.27
274	000094-75-7	2,4-dichloro-phenoxyacetic acid	1.7
275	000573-56-8	2,6-dinitrophenol	1.86
276	000095-57-8	2-chlorophenol	2.37
277	001420-07-1	2-tert-butyl-4,6-dinitrophenol	3.59
278	000609-19-8	3,4,5-trichlorophenol	4.23
279	000577-71-9	3,4-dinitrophenol	1.93
280	000591-35-5	3,5-dichlorophenol	3.46
281	002338-29-6	4,5,6,7-tetrachloro-2-(trifluoromethyl)-1H-benzimidazole	4.35
282	000609-93-8	4-methyl-2,6-dinitrophenol	2.26
283	006149-03-7	4-octylbenzene-1-sulfonate	3.63
284	004097-49-8	4-tert-butyl-2,6-dinitrophenol	3.23
285	002338-25-2	5,6-dichloro-2-(trifluoromethyl)-benzimidazole	3.05
286	000521-74-4	5,7-dibromo-8-hydroxyquinoline	3.35
287	000773-76-2	5,7-dichloro-8-hydroxyquinoline	2.7
288	016128-96-4	5-chloro-4-tert-butyl-2'-chloro-4'-nitrosalicylanilide	5.86
289	000130-16-5	5-chloro-8-hydroxyquinoline	3.16
290	000148-24-3	8-hydroxy-quinoline	2.16
291	000555-60-2	carbonyl cyanide m-chlorophenyl-hydrazone	4.05
292	000370-86-5	carbonyl cyanide p-methoxyphenylhydrazone	4.22
293	022494-42-4	diflunisal	2.73
294	000091-40-7	fenamic acid	2.28
295	000530-78-9	flufenamic acid	3.61
296	000124-07-2	octanoic acid	0.53
297	000608-71-9	pentabromo-phenol	5.02
298	000771-61-9	pentafluoro-phenol	1.74
299	001763-23-1	perfluorooctane-1-sulfonic acid	3.15
300	000335-67-1	perfluorooctanoic acid	2.34
301	001198-55-6	tetrachloro-catechol	2.74
302	004358-26-3	tetraphenylborate	5.2
303	000935-95-5	2,3,5,6-tetrachloro-phenol	3.49
304	020574-50-9	morantel	2
305	051384-51-1	metoprolol	1.28

306	000606-20-2	2,6-dinitro-toluene (2,6-DNT)	2.64

Table S2: Weighted stability analysis (stability_results.csv)

Fragment	Selection_Frequency
p1.O__	1
p1.___	1
p2.CB2CB_.4	1
p3.CA_CD3OD_.12	1
p3.CB1C__Cl_.41	1
p3.CB2CB2OA1.41	1
p3.CB2NA2CD_.11	1
s4.CB2Br1CB1CB1.144	1
s4.CB2CB1CB1CB2.441	1
p6.CA2CA2CA2CA2CA2CA2.11111	1
p4.CB1CB1CB2Cl1.441	1
p4.CB1CB2OA2CB2.411	1
p4.CB_CB_NB_CB_.444	1
p4.CA_CA_CA_OA_.111	1
p5.C__C__C__C__C__.1144	1
p4.CB1CB1CB2NA1.441	1
p3.C__C__OA1.11	1
p3.CA2CA2OA1.11	1
p6.CA1CB2CB1CB1CB2CA2.14441	1
p6.C__C__C__N__C__C__.14111	1
s5.CA4CA1CA1CA1CB2.1111	1
p2.CD3C__.2	0.999
s6.CB2CB2CB2CB2CB2CB2.14444	0.999
p3.NA1CD3OD_.12	0.999
p3.Cl1C__Cl1.11	0.999
p4.CB2CB2CB2CB2.444	0.999
p6.C__C__C__C__C__N__.14441	0.998
p5.CA1CA2CA2CA2CA2.1111	0.997
s6.CB2CB2CB1Cl1CB2OA1.44141	0.997
p2.CA1O__.1	0.995
s6.CB2CB2CA2CB1CB1OA1.41441	0.992
p5.Br1CB2CB2CB2CB2.1444	0.992
p4.CB1CB2CA3CD3.411	0.991
p3.C__CD3OA1.11	0.989
p3.CB_CA2N__.11	0.988
p3.CA4CA4CA_.11	0.987
p5.C__N__C__N__C__.1111	0.986
p2.N__N__.1	0.986
p2.CB_F__.1	0.985
p3.CA4CA_CD_.11	0.985
p4.CB1CB2CB2NA2.441	0.984
p5._____.4114	0.983
p6.CA1CA2CA2CA2CA2CA2.11111	0.976

p3.CA1CA_CA2.11	0.974
p6.CB1CB2CB1CB1CB2CB2.44444	0.973
p6.CB2CB2CB2OA2CB2CB2.44114	0.95
p3.CB2CB2S__.41	0.949
p2.CD2N__.1	0.9
p2.CD2NA3.1	0.847
p3.CA1C__CA2.11	0.755

Table S2: Weighted stability analysis: Top fragments (top_fragments.csv)

Fragment	Selection_Frequency
p1.O__	1
p1.___	1
p2.CB2CB_.4	1
p3.CB1C__Cl_.41	1
p3.CB2CB2OA1.41	1
p2.CD3C__.2	0.999
p2.CA1O__.1	0.995
p2.N__N__.1	0.986
p2.CB_F__.1	0.985
p3.CB2CB2S__.41	0.949
p2.CD2N__.1	0.9
p2.CD2NA3.1	0.847

Table S3: Mechanistic model coefficients (10-fragment model)

Fragment	Coefficient	Abs_Coefficient
p2.CB2CB_.4	1.159061	1.159061
p1.O__	-0.52736	0.527361
p1.___	0.396278	0.396278
p3.CB1C__Cl_.41	0.20617	0.20617
p3.Cl1C__Cl1.11	0.166204	0.166204
p2.CB_F__.1	-0.15091	0.150907
p4.CB1CB2CB2NA2.441	-0.14739	0.147391
p5.Br1CB2CB2CB2CB2.1444	0.101135	0.101135
p3.CB2CB2OA1.41	-0.05656	0.056565
p2.N__N__.1	0.051053	0.051053

Table S4: Statistical characterisation of mechanistic model coefficients (10-fragment model)

Fragment	Coefficient	Std_Error	CI_Lower_95	CI_Upper_95	p_value	Significant_p05	Significant_p01
p3.Cl1C__Cl1.11	9.04E-02	2.02E-02	3.45E-02	1.46E-01	1.09E-02	True	False
p1.O__	-4.89E-01	1.03E-02	-5.18E-01	-4.61E-01	1.19E-06	True	True

p3.CB2CB2OA1.41	-2.76E-02	1.05E-02	-5.68E-02	1.62E-03	5.86E-02	False	False
p2.CB2CB_.4	1.14E+00	1.24E-02	1.11E+00	1.18E+00	8.39E-08	True	True
p1.___	3.53E-01	2.03E-02	2.97E-01	4.09E-01	6.41E-05	True	True
p3.CB1C__Cl_.41	2.10E-01	1.42E-02	1.71E-01	2.49E-01	1.22E-04	True	True
p2.N__N__.1	7.03E-02	1.07E-02	4.05E-02	1.00E-01	2.82E-03	True	True
p4.CB1CB2CB2NA2.441	-8.53E-02	2.32E-02	-1.50E-01	-2.10E-02	2.12E-02	True	False
p2.CB_F__.1	-9.98E-02	2.28E-02	-1.63E-01	-3.66E-02	1.18E-02	True	False
p5.Br1CB2CB2CB2CB2.14 44	5.64E-02	1.86E-02	4.85E-03	1.08E-01	3.85E-02	True	False
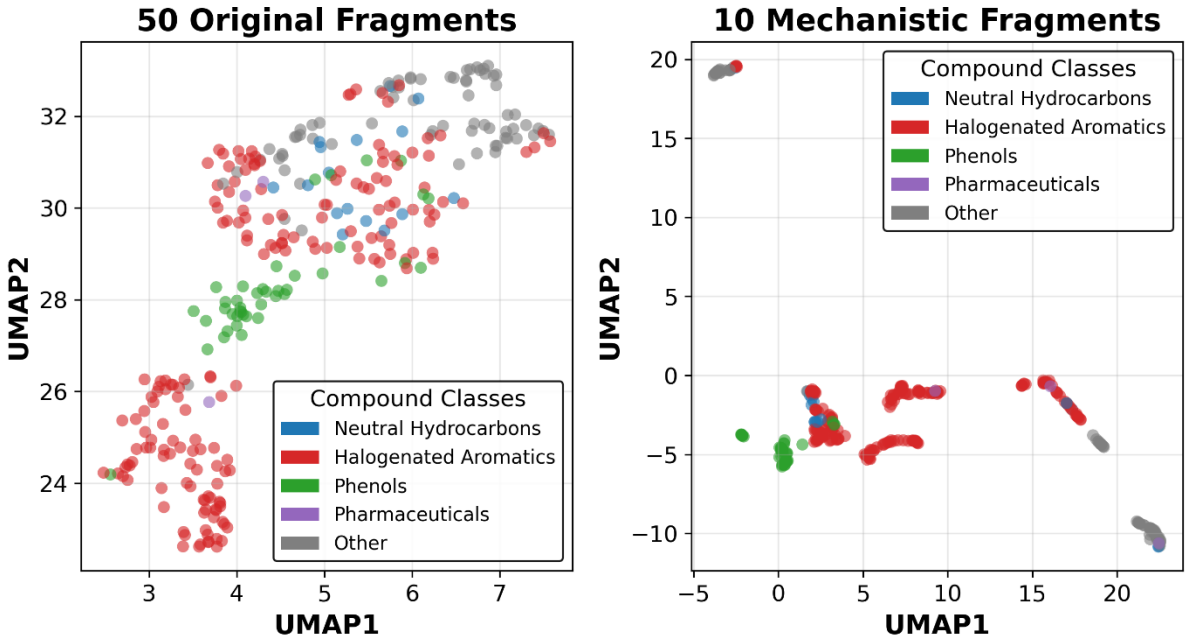
Figure S1: Comparison of UMAP diagrams for 50- and 10 fragment models